2 A speedboat of mass 450 kg is equipped with two identical motors. Each motor can produce 75 kW of power. When both motors are turned on, its maximum speed is 30 m s^{-1}. Assume that the drag force on the speedboat is proportional to the square of its speed.

(a) Determine the maximum speed of the boat when only one motor is turned on.

At maximum speed, driving force from motor = resistive force F.
Also $F = kv^2$, where k = constant.

Power supplied by motor $P = Fv_{max}$

$$= \left(kv_{max}^{\ 2}\right)v_{max}$$

$$= kv_{max}^{\ 3} \qquad [1]$$

$$\frac{P_1}{P_2} = \left(\frac{v_{max1}}{v_{max2}}\right)^3$$

$$v_{max1} = v_{max2} \times \sqrt[3]{\frac{P_1}{P_2}}$$

$$= 30 \times \sqrt[3]{0.5}$$

$$= 23.811$$

$$\approx 24 \text{ m s}^{-1} \qquad [1]$$

maximum speed = ... m s^{-1} [3]

(b) When both motors are turned on, the speedboat takes 9.0 s to reach the maximum speed of 30 m s^{-1}. Determine the amount of energy that is lost to the surroundings during that time.

Energy supplied by motor - energy lost = ke gained [1]

$$P_2 t - E_{lost} = \frac{mv^2}{2} - 0$$

$$E_{lost} = P_2 t - \frac{mv^2}{2}$$

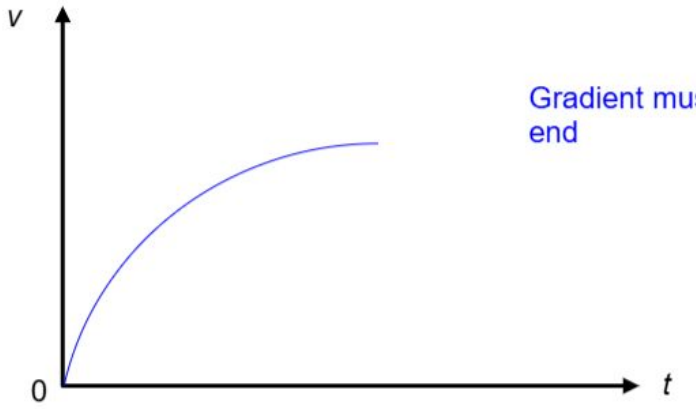
$$= \left(150000 \times 9.0\right) - \frac{450 \times 30^2}{2}$$

$$= 1.1475 \times 10^6 \text{ J}$$

$$\approx 1.1 \times 10^6 \text{ J} \qquad [1]$$

amount of energy lost = ... J [2]

(c) Sketch, on Fig. 2.1, the variation with time t of the speed of the speedboat v, until it reaches maximum speed.



Gradient must be zero at the end

Fig. 2.1

[1]

(d) When the speedboat is travelling at a speed of 20 m s^{-1}, both motors are turned off at the same time. The speedboat continues to travel an additional distance of 600 m before coming to rest. Determine the average resistive force acting on the speedboat.

Work done against resistive force = ke lost [1]

$$fx = \frac{mv^2}{2} - 0$$

$$\text{Average resistive force } f = \frac{mv^2}{2x}$$

$$= \frac{450 \times 20^2}{2 \times 600}$$

$$= 150 \text{ N} \quad [1]$$

average resistive force = ... N [2]

[Total: 8]